Volatility models for REMX: GARCH, GARCH-X, GARCHAND, GARCHND and EGARCH-X

Summary of all models estimated in the notebooks **03-GP-models-gaussian**, **03-GP-models-student** and **04-extension-models**. Each section contains the variance expression, a brief description of what the model is meant to demonstrate, and the main estimation results (Gaussian first, Student-t second). The *Sig.* column summarises significance: *** = 1%, ** = 5%, * = 10%.

1. GARCH(1,1) — Baseline

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1}$$

What we want to demonstrate: Canonical Bollerslev (1986) specification. It serves as the benchmark against which every sentiment-augmented model is compared. The goal is to confirm persistence ($\alpha+\beta$ close to 1) and that the basic parameters are significant; all later extensions are evaluated relative to this fit.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0857	0.0301	2.8497	0.0044	***
alpha[1]	0.0893	0.0161	5.5536	0.0000	***
beta[1]	0.8975	0.0182	49.2043	0.0000	***

```
Log-likelihood : -5982.3069
AIC : 11970.6138
BIC : 11988.3882
alpha[1] + beta[1] = 0.9867 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0878	0.0299	2.9308	0.0034	***
alpha[1]	0.0841	0.0150	5.5880	0.0000	***
beta[1]	0.9010	0.0178	50.6845	0.0000	***
nu	11.1674	2.3239	4.8054	0.0000	***

```
Log-likelihood : -5956.2330
AIC : 11920.4660
BIC : 11944.1652
alpha[1] + beta[1] = 0.9850 (stationary (<1))
nu = 11.1674 (smaller -> heavier tails; Gaussian as nu -> infinity)
```

2. GARCH-X (tone) — G&P Eq. (4)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot \text{tone}^2_{t-1}$$

What we want to demonstrate: Symmetric extension that injects the daily average news tone as an exogenous regressor in the variance equation. We want to test whether news intensity (regardless of sign) generates additional volatility in REMX, i.e. whether γ is statistically significant.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0835	0.0457	1.8258	0.0679	*
alpha	0.0905	0.0159	5.7033	0.0000	***
beta	0.8961	0.0179	50.0241	0.0000	***
gamma	0.0024	0.0213	0.1136	0.9096	

```
Log-likelihood : -5982.5250
AIC : 11973.0499
BIC : 11996.7491
alpha + beta : 0.9867 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1202	0.0460	2.6149	0.0089	***
alpha	0.0843	0.0153	5.5026	0.0000	***
beta	0.8984	0.0188	47.8456	0.0000	***
gamma	-0.0179	0.0155	-1.1549	0.2481	
nu	10.8940	2.2264	4.8932	0.0000	***

```
Log-likelihood : -5955.3892
AIC : 11920.7783
BIC : 11950.4023
alpha + beta : 0.9828 (stationary (<1))
Estimated degrees of freedom: nu_hat = 10.894
Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).
```

3. GARCH-X (log article growth) — G&P Eq. (4)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot \text{log\_art\_growth}^2_{t-1}$$

What we want to demonstrate: Same structure as the previous GARCH-X, but using the log-growth in article volume ($\log(1+N_t) - \log(1+N_{t-1})$) as a proxy for news intensity. The log transform replaces the raw growth rate to remove a single-day outlier (art_growth = 61 on 2015-10-23) that would otherwise dominate the squared term. We want to see whether sustained changes in media coverage (a significant γ) anticipate next-day volatility.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0757	0.0362	2.0947	0.0362	**
alpha	0.0910	0.0160	5.6763	0.0000	***
beta	0.8959	0.0181	49.5268	0.0000	***
gamma	0.0203	0.0372	0.5476	0.5840	

```
Log-likelihood : -5982.2558
AIC : 11972.5115
BIC : 11996.2107
alpha + beta : 0.9869 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0885	0.0359	2.4669	0.0136	**
alpha	0.0852	0.0150	5.6883	0.0000	***
beta	0.8996	0.0176	51.0511	0.0000	***
gamma	0.0012	0.0356	0.0346	0.9724	
nu	11.1206	2.3100	4.8142	0.0000	***

```
Log-likelihood : -5956.3176
AIC : 11922.6352
BIC : 11952.2592
alpha + beta : 0.9848 (stationary (<1))
Estimated degrees of freedom: nu_hat = 11.121
Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).
```

4. GARCHAND (tone) — G&P Eq. (5)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma\,(1 + d_1 \cdot \theta) \cdot \mathrm{tone}^2_{t-1},\ d_1 = 1\{\mathrm{tone}_{t-1} < 0\}$$

What we want to demonstrate: Asymmetric version of GARCH-X: when the previous day's tone is negative, its impact on variance is amplified by the factor $(1 + \theta)$. We want to demonstrate that negative news drives a stronger volatility response than positive news ($\theta > 0$ and significant).

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0899	0.0353	2.5503	0.0108	**
alpha	0.0904	0.0159	5.6758	0.0000	***
beta	0.8958	0.0180	49.7159	0.0000	***
gamma	-0.0009	0.0041	-0.2274	0.8201	
theta	0.4979	6.7407	0.0739	0.9411	

```
Log-likelihood : -5982.5410
AIC : 11975.0819
BIC : 12004.7059
alpha + beta : 0.9862 (stationary (<1))
Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.481
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1284	0.0475	2.7003	0.0069	***
alpha	0.0839	0.0159	5.2868	0.0000	***
beta	0.8966	0.0201	44.5784	0.0000	***
gamma	-0.0005	0.0002	-2.5306	0.0114	**
theta	47.9057	16.0878	2.9778	0.0029	***
nu	10.9218	2.2666	4.8185	0.0000	***

```
Log-likelihood : -5954.5020
AIC : 11921.0040
BIC : 11956.5528
alpha + beta : 0.9805 (stationary (<1))
Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.481
Estimated degrees of freedom: nu_hat = 10.922
```

5. GARCHAND (log article growth) — G&P Eq. (6)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot d_2 \cdot \text{log\_art\_growth}^2_{t-1},\ d_2 = 1\{\text{log\_art\_growth}_{t-1} > 0\}$$

What we want to demonstrate: The article-volume effect is switched on only on days in which the news flow GROWS relative to the previous day (log_art_growth > 0). We want to show that novelty (rising coverage) is what moves volatility, rather than low news volumes.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0730	0.0356	2.0533	0.0400	**
alpha	0.0903	0.0159	5.6646	0.0000	***
beta	0.8964	0.0181	49.4520	0.0000	***
gamma	0.0521	0.0736	0.7077	0.4791	

```
Log-likelihood : -5982.0589
AIC : 11972.1179
BIC : 11995.8171
alpha + beta : 0.9867 (stationary (<1))
Fraction of days with log_art_growth_{t-1} > 0 (d2 = 1): 0.476
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0842	0.0351	2.4029	0.0163	**
alpha	0.0851	0.0149	5.7110	0.0000	***
beta	0.8998	0.0176	51.0085	0.0000	***
gamma	0.0177	0.0677	0.2610	0.7941	
nu	11.1560	2.3245	4.7994	0.0000	***

```
Log-likelihood : -5956.2750
AIC : 11922.5499
BIC : 11952.1739
alpha + beta : 0.9849 (stationary (<1))
Fraction of days with log_art_growth_{t-1} > 0 (d2 = 1): 0.476
Estimated degrees of freedom: nu_hat = 11.156
```

6. GARCHND (tone & log article growth) — G&P Eq. (7)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot d_3 \cdot x^2_{t-1},\ d_3 = 1\{\sigma^2_{t-1} \geq \kappa\},\ x \in \{\text{tone, log\_art\_growth}\}$$

What we want to demonstrate: The news impact activates only when the previous day's variance exceeds a threshold κ, calibrated from an annualised volatility level (30% and 50%). We want to test whether the sentiment effect is non-linear in the volatility regime: a significant γ in the high-volatility regime would imply that news amplifies volatility only when the market is already stressed.

Gaussian

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0967	0.0247	3.9157	0.0001	***
alpha	0.0888	0.0101	8.7465	0.0000	***
beta	0.8891	0.0139	63.7458	0.0000	***
gamma	0.0375	0.0342	1.0965	0.2729	

```
Log-likelihood : -5979.6159
AIC : 11967.2319
BIC : 11990.9311
alpha + beta : 0.9779 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.616
```

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0532	0.0115	4.6104	0.0000	***
alpha	0.0872	0.0027	32.6900	0.0000	***
beta	0.9101	0.0032	287.2016	0.0000	***
gamma	-0.4811	0.0227	-21.2193	0.0000	***

```
Log-likelihood : -5977.0528
AIC : 11962.1055
BIC : 11985.8047
alpha + beta : 0.9973 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.072
```

GARCHND x = log_art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1088	0.0216	5.0458	0.0000	***
alpha	0.0945	0.0309	3.0598	0.0022	***
beta	0.8795	0.0116	75.8273	0.0000	***
gamma	0.1299	0.4061	0.3200	0.7490	

```
Log-likelihood : -5979.8694
AIC : 11967.7387
```

```
BIC : 11991.4379
alpha + beta : 0.9740 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.621
```

GARCHND x = log_art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0551	0.0134	4.1159	0.0000	***
alpha	0.0901	0.0014	65.0193	0.0000	***
beta	0.9077	0.0003	2997.4667	0.0000	***
gamma	-1.6273	0.0741	-21.9478	0.0000	***

```
Log-likelihood : -5975.4760
AIC : 11958.9520
BIC : 11982.6512
alpha + beta : 0.9978 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.066
```

Student-t

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0918	0.0282	3.2498	0.0012	***
alpha	0.0850	0.0137	6.2072	0.0000	***
beta	0.8981	0.0162	55.5481	0.0000	***
gamma	0.0081	0.0239	0.3400	0.7339	
nu	11.2155	2.2412	5.0041	0.0000	***

```
Log-likelihood : -5956.2451
AIC : 11922.4903
BIC : 11952.1143
alpha + beta : 0.9830 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.613
Estimated degrees of freedom: nu_hat = 11.215
```

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0481	0.0174	2.7651	0.0057	***
alpha	0.0810	0.0047	17.3331	0.0000	***
beta	0.9165	0.0078	116.7886	0.0000	***
gamma	-0.7553	0.0470	-16.0863	0.0000	***
nu	11.5147	2.3839	4.8302	0.0000	***

```
Log-likelihood : -5952.0998
AIC : 11914.1995
BIC : 11943.8235
```

```
alpha + beta : 0.9975 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.062
Estimated degrees of freedom: nu_hat = 11.515
```

GARCHND x = log_art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1010	0.0299	3.3781	0.0007	***
alpha	0.0877	0.0126	6.9575	0.0000	***
beta	0.8915	0.0163	54.5380	0.0000	***
gamma	0.0620	0.0480	1.2913	0.1966	
nu	11.2835	2.2238	5.0740	0.0000	***

```
Log-likelihood : -5955.7018
AIC : 11921.4036
BIC : 11951.0276
alpha + beta : 0.9793 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.621
Estimated degrees of freedom: nu_hat = 11.283
```

GARCHND x = log_art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0479	0.0291	1.6458	0.0998	*
alpha	0.0800	0.0107	7.5076	0.0000	***
beta	0.9176	0.0160	57.3023	0.0000	***
gamma	-1.3931	0.2035	-6.8442	0.0000	***
nu	11.3272	2.3644	4.7907	0.0000	***

```
Log-likelihood : -5951.2465
AIC : 11912.4931
BIC : 11942.1171
alpha + beta : 0.9976 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.063
Estimated degrees of freedom: nu_hat = 11.327
```

7. EGARCH(1,1)-X — Extension model

Conditional variance:

$$\ln(\sigma^2_t) = \omega + \alpha\,[|z_{t-1}| - E|z_{t-1}|] + \xi \cdot z_{t-1} + \beta \cdot \ln(\sigma^2_{t-1}) + \gamma_1 \cdot \text{tone}_{t-1} + \gamma_2 \cdot \text{log\_art\_growth}_{t-1}$$

What we want to demonstrate: Log-variance specification (Nelson, 1991) with two exogenous regressors. It (i) guarantees positivity of σ^2 without parameter restrictions, (ii) captures asymmetry through the leverage term ξ (negative shocks weigh more than positive ones), and (iii) allows tone and log_art_growth to enter linearly (not as squares). We want to show that EGARCH-X captures asymmetry more cleanly than GARCHAND, and to test whether γ_1 and γ_2 are significant in log-variance rather than in squared variance.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.1007	0.0375	-2.6841	0.0073	***
omega	0.0342	0.0146	2.3521	0.0187	**
alpha	0.3033	0.0625	4.8542	0.0000	***
xi	-0.0197	0.0168	-1.1735	0.2406	
beta	0.9724	0.0139	69.8722	0.0000	***
gamma_tone	-0.0007	0.0098	-0.0685	0.9454	
gamma_log_artg	0.0843	0.0683	1.2342	0.2171	

```
Log-likelihood : -6011.7360
AIC : 12037.4719
BIC : 12078.9455
beta = 0.9724 (stationary (|beta|<1))
Half-life of a log-variance shock: 24.8 trading days
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.0177	0.0025	-6.9528	0.0000	***
omega	0.0319	0.0102	3.1320	0.0017	***
alpha	0.1565	0.0253	6.1860	0.0000	***
xi	-0.0143	0.0107	-1.3360	0.1816	
beta	0.9792	0.0065	150.4773	0.0000	***
gamma_tone	0.0106	0.0049	2.1500	0.0316	**
gamma_log_artg	-0.0099	0.0601	-0.1645	0.8693	
nu	10.7946	2.1052	5.1275	0.0000	***

```
Log-likelihood : -5951.4540
AIC : 11918.9080
BIC : 11966.3063
beta = 0.9792 (stationary (|beta|<1))
Half-life of a log-variance shock: 33.0 trading days
```

nu = 10.7946 (Student-t dof; lower -> heavier tails. Note: the t-ratio for nu tests H0: nu = 0, which is not the meaningful null. To test Gaussian vs Student-t use an LR test against the Gaussian EGARCH-X.)